

STUDY QUESTIONS 3

1. **A warehouse manager-** A warehouse manager needs to decide which products to stock for the upcoming season. The warehouse has two critical limitations: limited floor space (measured in square meters) and a maximum load-bearing capacity for the shelving system (measured in kilograms).

Each product j is characterized by three attributes: expected profit margin (profit_j , in TL), floor space required (space_j , in square meters), and weight (weight_j , in kilograms). The warehouse has a total available floor space of S square meters and a maximum weight capacity of W kilograms for its shelving system.

The manager must decide which products to stock to maximize total profit while respecting both the floor space limitation and the weight capacity of the shelving infrastructure.

- Define appropriate decision variables for this problem.
- Write the complete mathematical formulation including: (i) Objective function, (ii) Floor space constraint, (iii) Weight capacity constraint, and (iv) Variable type constraints.
- Suppose there are 4 products under consideration with the following characteristics. Write the specific mathematical model for this problem instance using the data below.

Product	Profit (TL)	Floor Space (m ²)	Weight (kg)
A	800	12	150
B	650	8	200
C	900	15	120
D	550	6	180

Available floor space: $S = 30$ square meters, Weight capacity: $W = 400$ kilograms

2. **Factory Production Planning Problem-** SteelCars Factory, which produces automobiles and bicycles, wants to optimize its monthly production plan. The factory manufactures passenger cars and bicycles, but resources are limited, and there are specific market demands. The table below shows these products' resource usage and their profits per unit:

Product	Profit per Unit	Steel per Unit	Labor per Unit	Assembly per Unit
Car	2,500 TL	1,500 g	30 hours	20 hours
Bicycle	120 TL	20 kg	2 hours	3 hours

In addition, the factory has limited available resources. The maximum amount of steel available is 30,000 kg, the total available labor time is 2,400 hours, and the total available assembly time is 1,800 hours. Besides these resource limitations, the factory also sets monthly production and sales requirements: no more than 15 cars can be sold, and at least 50 bicycles must be produced.

- Formulate a linear programming model for this project without solving it for efficient solution. Clearly define decision variables, constraints, objective function and parameters.

3. **Production Scheduling in a Sweet Factory-** Ayşe is the production manager of a small dessert factory in Istanbul that produces boxes of chocolates and cookies. The next two months are expected to be especially busy because a major supermarket chain has placed a large order for an upcoming promotional campaign. Therefore, Ayşe needs to plan production carefully over the two-month period in order to meet the required delivery deadlines without exceeding the factory's monthly production capacity.

The factory has a maximum production capacity of 600 hours per month. Producing one box of chocolates requires 4 hours, while producing one box of cookies requires 2 hours. The supermarket requires at least 100 boxes of cookies to be delivered by the end of the first month and at least 150 boxes of chocolates to be delivered by the end of the second month. Products produced in the first month can be stored and used to meet future delivery requirements.

Ayşe wants to operate as efficiently as possible by minimizing the total production time used over the two months.

- a) Formulate a linear programming model that helps Ayşe determine how many boxes of chocolates and cookies should be produced in each month in order to meet the delivery requirements with minimum total production time. Clearly define decision variables, constraints, objective function and parameters.
4. **Preparing a Healthy Feed Mix-** Emily is a nutrition specialist working for a large cattle farm. Each day, she must prepare a livestock feed mix that satisfies the animals' nutritional requirements at the lowest possible cost. The farm has access to three feed ingredients: corn, soybean meal, and hay. Each ingredient has a different cost and provides different amounts of protein, fiber, and energy per kilogram.

The daily feed mix must contain at least 50 units of protein, at least 30 units of fiber, and at least 180 units of energy. Emily's goal is to find the cheapest possible combination of the three ingredients while satisfying all of these nutritional requirements.

The nutritional content and cost of each ingredient are given in the table below:

Ingredient	Cost per kg (\$)	Protein (units/kg)	Fiber (units/kg)	Energy (units/kg)
Corn	0.50	4	2	20
Soybean Meal	0.80	10	1	15
Hay	0.30	2	5	10

- a) Formulate a linear programming model to help Emily determine how many kilograms of each ingredient should be used per day in order to minimize the total cost while ensuring that the feed mix meets or exceeds the required levels of protein, fiber, and energy. Clearly define decision variables, constraints, objective function and parameters.
5. **Steel Rod Cutting-** Jake is the operations manager at a metal workshop that produces steel rods of various lengths for construction companies. Each day, he receives standard steel rods that are 100 inches long and must decide how to cut them into smaller pieces to fulfill customer orders. The workshop offers three standard sizes for sale: 20-inch, 30-inch, and 50-inch rods, each generating a different amount of profit.
- Jake's goal is to determine how to cut each 100-inch rod into a combination of these standard sizes in order to maximize total profit while meeting customer demand for each rod size.
- The current order requires at least 80 pieces of 20-inch rods, at least 60 pieces of 30-inch rods, and at least 40 pieces of 50-inch rods. Each 100-inch rod can be cut using one of five possible cutting patterns. The first pattern produces five pieces of 20-inch rods. The second pattern produces three pieces of 30-inch rods and leaves 10 inches of waste. The third pattern produces two pieces of 50-inch rods. The fourth pattern produces one 50-inch rod, one 30-inch rod, and one 20-inch rod. The fifth pattern produces one 50-inch rod and two 20-inch rods, leaving 10 inches of waste.
- Each rod size also yields a different profit. A 20-inch rod earns \$4 per piece, a 30-inch rod earns \$6 per piece, and a 50-inch rod earns \$10 per piece.
- Jake wants to decide how many 100-inch rods should be cut using each possible pattern in order to meet the order requirements and maximize total profit.
- a) Formulate a linear programming model for this project without solving it, for an efficient solution. Clearly define the decision variables, constraints, objective function, and parameters.

6. **Staff Assignment at a Summer Camp-** Sophia is the coordinator of a summer camp that organizes weekly outdoor activities for children. Each week, the camp offers three main activities: swimming, arts and crafts, and hiking. Sophia has a team of six staff members, and each staff member has different skills and preferences for these activities. Some staff members are strong swimmers, some are creative and good at arts and crafts, and others are experienced hikers.
- Each activity requires two staff members to supervise it. To provide a safe and enjoyable environment for the children, Sophia wants to assign staff members to activities in a way that maximizes the overall skill score. She has rated each staff member's skill level for each activity on a scale from 1 to 10, where a higher score indicates a better match for that activity. The skill scores are given in the table below:

Staff	Swimming	Arts & Crafts	Hiking
Alice	9	6	5
Ben	7	8	6
Chloe	6	9	8
David	10	5	7
Emma	5	7	9
Frank	8	6	10

Due to time constraints, each staff member can be assigned to at most one activity per week. In addition, each activity must be supervised by exactly two staff members.

- a) Formulate a linear programming model to help Sophia assign staff members to activities in a way that maximizes the total skill score across all assignments. Clearly define the decision variables, objective function, and constraints. You do not need to solve the model.

7. **Assigning Technicians-** A large electronics service company has received four urgent repair requests from different clients across the city. The company has four available technicians: John, Maria, Kevin, and Sophie. Each technician has a different level of expertise and familiarity with each type of repair job, which affects the time required to complete the task.

The manager, Lisa, wants to assign technicians to repair jobs as efficiently as possible. Each technician must be assigned to exactly one repair job, and each repair job must be handled by exactly one technician. The goal is to minimize the total repair time required to complete all four jobs.

Lisa has estimated the time, in hours, that each technician would need to complete each job. The estimated times are given in the table below:

Technician	Job A	Job B	Job C	Job D
John	3	5	2	7
Maria	6	4	3	5
Kevin	5	3	6	2
Sophie	4	6	5	3

- a) Formulate a linear programming model to determine the optimal assignment of technicians to jobs in order to minimize the total time spent. Clearly define the decision variables, objective function, and constraints. You do not need to solve the model.

8. **Laptop Transportation-** A company operates two factories, one in Chicago and one in Dallas, that produce laptop computers. These laptops must be delivered to three warehouses located in New York, Atlanta, and Denver. Each factory has a limited supply of laptops, and each warehouse has a specific demand that must be satisfied.

The transportation cost per laptop differs depending on the factory and warehouse combination because of distance and shipping methods. The company wants to determine the most cost-effective way to ship laptops from the factories to the warehouses, while ensuring that each warehouse's demand is met and no factory ships more than its available supply.

The unit transportation costs, factory supplies, and warehouse demands are given in the table below:

Factory	New York	Atlanta	Denver	Supply
Chicago	\$4	\$6	\$8	500
Dallas	\$5	\$4	\$3	400
Demand	300	350	250	

- a) Formulate a linear programming model for this transportation problem without solving it. Clearly define the decision variables, objective function, and constraints.

9. **Emma's Investment Portfolio-** Emma is a financial analyst with \$1,000,000 to invest for one of her clients. She is considering five different investment options: government bonds, real estate fund, tech stocks, international index fund, and a high-risk start-up portfolio. Each investment option has a different expected annual return, and Emma's goal is to maximize the total expected return.

However, Emma must also follow several investment requirements. No more than 40% of the total investment can be allocated to tech stocks and the high-risk start-up portfolio combined. In addition, at least 20% of the total investment must be placed in government bonds. Emma must allocate the entire \$1,000,000 budget among the available investment options.

The expected annual returns for each investment option are given in the table below:

Investment Option	Expected Return (%)
Government Bonds	4
Real Estate Fund	6
Tech Stocks	10
International Index Fund	7
High-Risk Start-up Portfolio	15

- a) Formulate a linear programming model to help Emma determine how much money should be invested in each option in order to maximize the total expected return under the given constraints. Clearly define the decision variables, objective function, and constraints. You do not need to solve the model.

10. **CPU Production in AI Race-** As the AI models gets more complex, these models require more computation power. Nowadays, CPU producers get more attention in the market. Your electronics company wants to enter this market with a new monthly project named K-CPU. In this project you are tasked with creating a mathematical model for optimizing the following:

In K-CPU project there is three CPU prototypes: Type A, Type B and Type C. To build those CPU's, machinery require silicon, copper and aluminum. For this project, management allocated 4500 kilograms of silicon, 8000 kilograms of copper and 5000 kilograms of aluminum.

Due to its efficient and integrated neural processing unit, one unit of Type A CPU requires 4 kilograms of silicon, 5 kilograms of copper and 4 kilograms of aluminum. On the other hand, one unit Type B CPU requires 2 kilograms of silicon, 8 kilograms of copper and 3 kilograms of aluminum due to its size. Lastly, during production of Type C CPU's requires different alloys of aluminum thus one unit of Type C CPU require 3 kilograms of silicon, 6 kilograms of copper and 6 kilograms of aluminum.

During CPU production, machinery require water to cool down. Each one unit of CPU types A, B, and C consumes 100 liters, 60 liters, and 85 liters of water, respectively. Additionally, during production liquid and gas wastes are produced as a byproduct. Each one unit of CPU types A, B, and C generates 3 liters, 1 liter, and 1.5 liters of liquid waste, respectively. To process this liquid waste, 20 liters of water is used in dilution process for each liter of liquid waste.

Lastly, production of each one unit of CPU types A, B, and C produces $45 m^3$, $25 m^3$, and $30 m^3$ of gas waste, respectively.

Your company has a sustainability goal to reduce water consumption and gas emissions. Thus, management team has set a water consumption limit of 350.000 liters and gas emission limit of $65.000 m^3$ for this project.

Finally, management team has determined the market prices of each product as the following: one unit of Type A CPU is 9500\$, one unit of Type B CPU is 5500\$, and one unit of Type C CPU is 7000\$. Management aims to obtain highest total revenue possible under given restrictions.

- a) Formulate a linear programming model for this project without solving it for efficient solution. Clearly define decision variables, constraints, objective function and parameters.

11. **EBG Logistics' Transportation Problem-** One of the leading logistics company, EBG Logistics have reached out to you about their transportation problem. The company wants you to optimize their shipments from warehouses to stores. The management team provided you following tables:

Table 1 - Supply	
Warehouses	X_i = Available Amount of Items
W_1	25
W_2	60
W_3	42
W_4	34
W_5	55

Table 2 – Demand	
Stores	Y_j = Required Amount of Items
S_1	37
S_2	22
S_3	28
S_4	55
S_5	24

Cost of Transportation per Item (C_{ij}) (\$ / per item)					
	S_1	S_2	S_3	S_4	S_5
W_1	12	21	26	23	30
W_2	15	19	29	28	28
W_3	13	17	20	25	25
W_4	10	13	18	23	27
W_5	11	25	24	10	23

Due to the fact that two of company's warehouses, W_3 and W_4 , are in neighboring country; company must pay customs fee for each item imported from that country. (Remaining warehouses and stores are in home country). Customs fee is 8\$ per item.

Management team's goal is to minimize transportation cost while satisfying given demands of stores.

- a) Formulate a linear programming model for this project without solving it for efficient solution. Clearly define decision variables, constraints, objective function and parameters.